Topic 12: Aromatic Compounds

Hydrocarbons II: benzene, naming, EAS, directors, phenols

CHEMISTRY (PROF. L. RANAY) | LECTURE QUIZ, NOV 7: TOPICS 12-14

What she stressed for the quiz

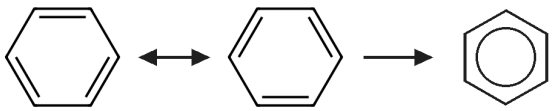
Meta-nitrotoluene: **nitration first**, then $CH_3Cl/AlCl_3$. Know **EDG vs EWG** without a table. **o/p director = 2 products, meta = 1.** Aromatics are judged by the **substituent**, not the ring. **ALL hydrocarbons (benzene too) are insoluble in water.**

Quiz format

I modified true/false. **II** naming (name it or draw it). **III** predict the products. **IV** arrange boiling points, predict solubility.

BENZENE

- **C_6H_6**, simplest **arene**. Named "aromatic" from smelly derivatives (**benzaldehyde** almonds, **vanillin**, **cinnamaldehyde**).
- **Faraday** isolated it, **Kekulé** proposed the ring (snake biting its tail), **Pauling** gave **resonance**.
- Real benzene is one **resonance hybrid**: all 6 C–C bonds equal, between single and double. The π electrons are **delocalized** (circle in the hexagon), so you cannot number the double bonds.



Two Kekulé structures (arrow ↔) = one hybrid with equal bonds

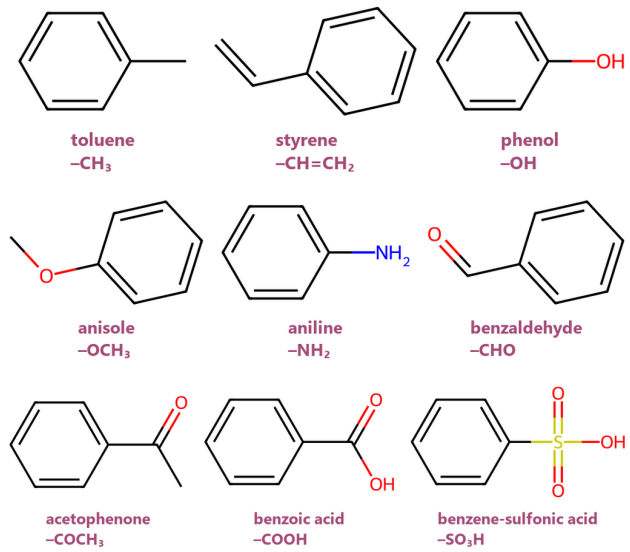
- **Not an alkene** because its π electrons are delocalized, **not** "because it is cyclic" (cyclohexene is cyclic). It does **substitution**, not addition (no reaction with cold $KMnO_4$ or Br_2 alone).
- Every benzene reaction needs a **catalyst** ($FeCl_3$, $FeBr_3$, H_2SO_4, $AlCl_3$).

Note: Benzene has **4** degrees of unsaturation (3 π bonds + 1 ring), not 3.

NAMING AROMATICS

MONOSUBSTITUTED

- No special name: substituent + **benzene**, one word, **no locant**: **chlorobenzene**, **nitrobenzene**, **ethylbenzene**.
- These 9 replace "benzene" as the parent name:

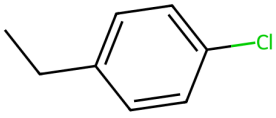


Retained parent names (memorize the 9)

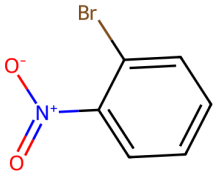
- **Phenyl (Ph–)** = the ring as a branch: **4-phenyl-1-butene**.

TWO GROUPS: ORTHO, META, PARA

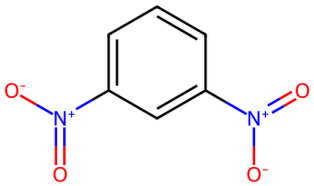
- **1,2 = ortho**, **1,3 = meta**, **1,4 = para** (only for TWO groups).
- No retained name: parent **benzene**, lowest numbers, tie goes to **alphabetical**. A retained-name group is **C1** (no "1" written).
- Two CH_3 = **xylene**, never "di-toluene".



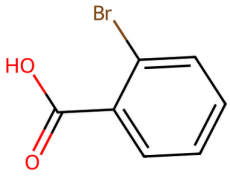
1-chloro-4-ethylbenzene
(p-): no retained name, alphabetical



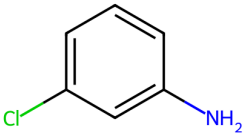
1-bromo-2-nitrobenzene
(o-)



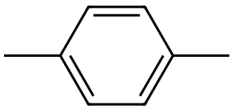
1,3-dinitrobenzene
(m-)



2-bromobenzoic acid
(retained name = C1)



3-chloroaniline
(m-chloroaniline)

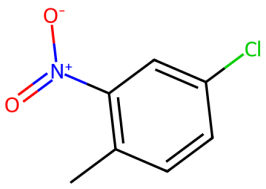


p-xylene
(1,4-dimethylbenzene)

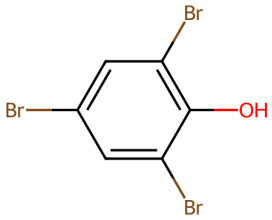
Naming examples: two substituents

THREE OR MORE GROUPS: NUMBERS ONLY

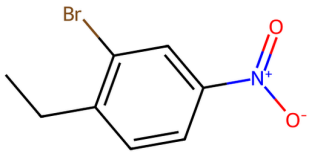
- Never o/m/p. Retained group = parent and C1; else benzene with the **lowest set**, then **alphabetical**.
- Several retained groups: highest priority wins: **acid > aldehyde > ketone > alcohol > amine > ether/halide/nitro**.



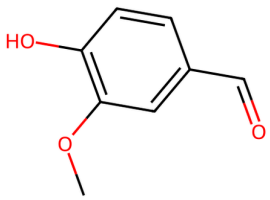
4-chloro-2-nitrotoluene



2,4,6-tribromophenol



2-bromo-1-ethyl-4-nitrobenzene
(no retained name: lowest set 1,2,4)



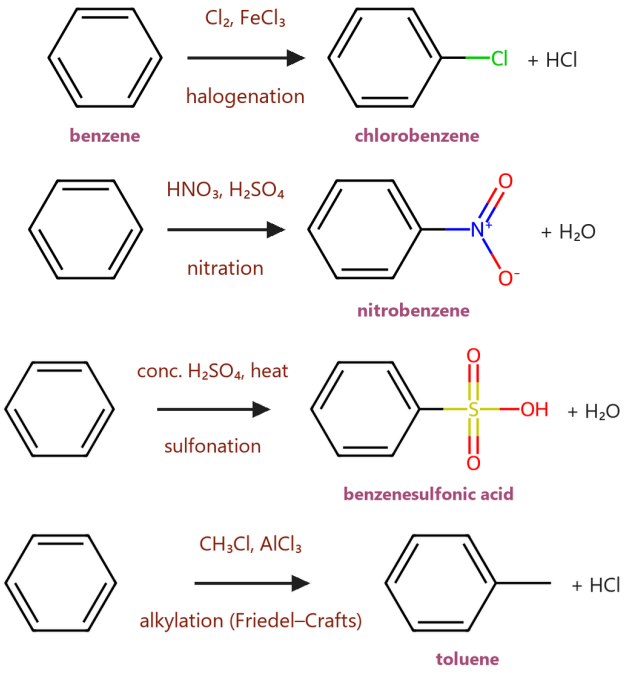
vanillin =
4-hydroxy-3-methoxybenzaldehyde
(aldehyde wins)

Naming examples: three or more groups

ELECTROPHILIC AROMATIC SUBSTITUTION (EAS)

- **Electrophile** (Cl^+, Br^+, NO_2^+, SO_3H^+, R^+) = Lewis acid. The **ring** is the nucleophile (Lewis base). Ionic, **not** a radical reaction like alkanes.
- Mechanism (3 steps, halogenation): **(1)** Cl_2 + $FeCl_3$ gives **Cl^+** + $FeCl_4^-$. **(2)** ring π electrons attack Cl^+, giving a resonance-stabilized carbocation (C holds H and Cl). **(3)** $FeCl_4^-$ removes H^+, ring restored, **HCl** + **$FeCl_3$** regenerated.

Electrophile: Cl^+ (Br^+ with $FeBr_3$), NO_2^+, SO_3H^+, R^+. Halogenation and alkylation make HX; nitration and sulfonation make H_2O.

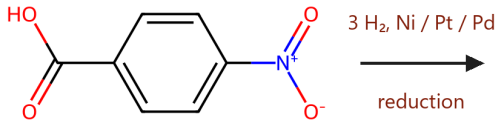


The four EAS reactions: reagent over the arrow

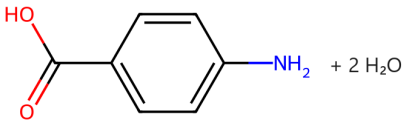
- Use **pure Cl_2 or Br_2**, not HCl or HBr.
- **By-product:** halogenation and alkylation make **HX**; nitration and sulfonation make **H_2O**.
- Fuming sulfonation = H_2SO_4 + SO_3. Detergent (SDS): dodecylbenzene + H_2SO_4 gives the sulfonic acid; + NaOH gives **sodium dodecylbenzenesulfonate** (water-soluble).

NITRO REDUCTION AND SIDE-CHAIN OXIDATION

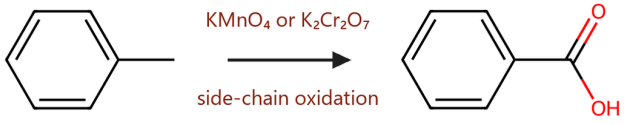
Only $-NO_2$ becomes $-NH_2$ (ring and –COOH untouched). The ring never oxidizes; the alkyl side chain becomes –COOH.



4-nitrobenzoic acid



4-aminobenzoic acid (PABA)



toluene or ethylbenzene — benzoic acid

$-NO_2$ becomes $-NH_2$; alkyl side chain becomes –COOH

- Reduction = gain of H or loss of O. Oxidation = gain of O or loss of H.

DIRECTORS: ORTHO/PARA VS META

Spot the director without the table

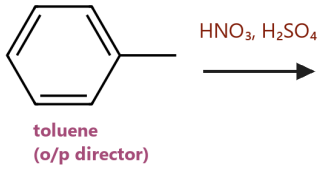
EDG: ortho / para	EWG: meta
$-NH_2$, $-NHR$, $-NR_2$ –OH, –OR $-R$ (alkyl), $-C_6H_5$ single bonds only, no C=O, not many O Faster than benzene: write 2 products	C=O on the ring atom: –CHO, –COR, –COOH, –COOR two or more O: $-SO_3H$, $-NO_2$ (slowest) –CN, $-NR_3^+$ Slower than benzene: write 1 product

Halogens (–F –Cl –Br –I) are the exception: they slow the ring (EWG) but still direct ortho / para.

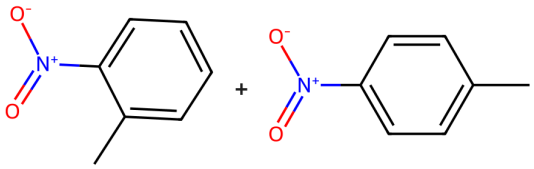
EDG versus EWG

- Group **already on the ring** decides where the next group goes and the speed.
- **EDG** (activating, o/p): $-NH_2$, –OH, –OR, $-R$, $-C_6H_5$. **EWG** (deactivating, meta): –CHO, –COR, –COOH, –COOR, $-SO_3H$, –CN, $-NR_3^+$, $-NO_2$ (slowest).
- **Halogens**: slow the ring yet **o/p** (write 2 products).

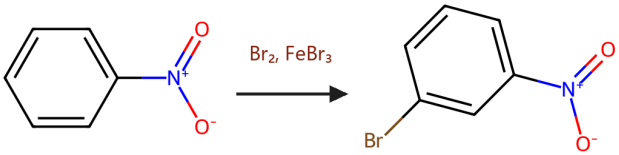
o/p director: write TWO products. meta director: write ONE.



toluene (o/p director)



o-nitrotoluene — p-nitrotoluene



nitrobenzene (meta director) — m-bromo-nitrobenzene ONE product only

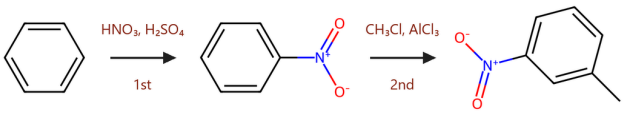
Predicting products: two for o/p, one for meta

HER FLAGGED ITEM

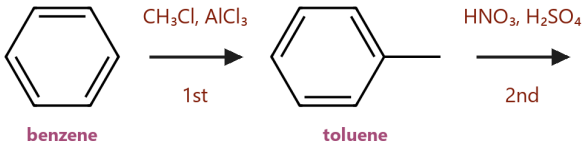
Which reaction goes FIRST to make meta-nitrotoluene?

Nitration first (HNO_3/H_2SO_4), then alkylation ($CH_3Cl/AlCl_3$). $-NO_2$ is a meta director. Alkylation first gives o- and p-nitrotoluene.

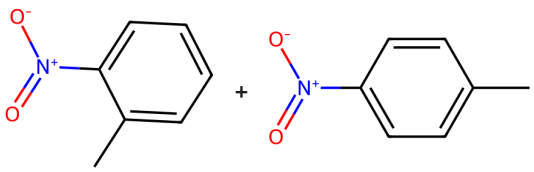
Her flagged item: meta-nitrotoluene needs NITRATION FIRST (the group already on the ring decides).



benzene — nitrobenzene — m-nitrotoluene



benzene — toluene



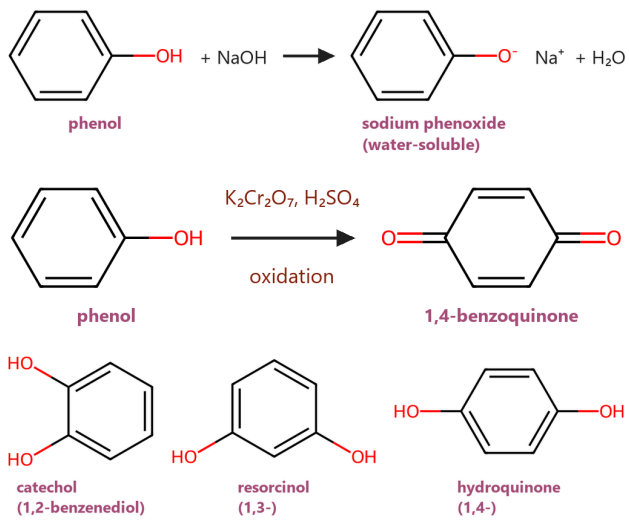
o-nitrotoluene — p-nitrotoluene

Order of steps decides the isomer

PHENOLS

- **–OH on a ring carbon**; a **separate family, not an alcohol** (benzyl alcohol has the OH on a CH_2).

- **Weak acid (pK_a about 10)**: insoluble in water, but **NaOH** makes soluble **sodium phenoxide**.
- **Cresols** = methylphenols. **Catechol** (1,2), **resorcinol** (1,3), **hydroquinone** (1,4).
- Oxidation ($K_2Cr_2O_7$) gives **1,4-benzoquinone** (CoQ, electron carriers).



Phenol: acid-base, oxidation, and the three benzenediols

- Natural phenols: **capsaicin** (chili), **thymol** (mouthwash, Lysol), **vanillin**, **urushiol** (poison ivy), **BHA / BHT** (food antioxidants).

END OF LESSON

Topic 13: Alcohols, Amines, Ethers, Thiols

Naming, boiling point, solubility, reactions

CHEMISTRY (PROF. L. RANAY) | LECTURE QUIZ, NOV 7: TOPICS 12-14

What she stressed for the quiz

Essay trick: ALL hydrocarbons and thiols are insoluble. Amine + acid = **salt only, "walay water!"** NMAT: **arrange boiling points**, **Markovnikov** (addition) vs **Zaitsev** (elimination), **$KMnO_4$, $K_2Cr_2O_7$, PCC**. **Not on the quiz:** cyclic-ether naming (know THF, ethylene oxide) and ether reactions.

BOILING POINT AND SOLUBILITY RULES

BOILING POINT: low to high (similar size)

Family	Why
1. Hydrocarbons	London dispersion only
2. Thiols	S–H barely polar, almost no H-bonding
3. Ethers	slightly polar, no O–H
4. Aldehydes / ketones	C=O dipole–dipole, no H-bonding
5. Amines	N–H···N bonds (weaker than O–H; 3° amines have none)
6. Alcohols	O–H···O hydrogen bonds
7. Carboxylic acids	dimers (double H-bond)

Same family: more carbons = HIGHER bp; more branching = LOWER bp (less surface). Order: IMF first, then size, branching last.

Boiling point order

WATER SOLUBILITY (her carbon boundaries)

Family	C1–C3	C4–C5	C6+
Alcohols, amines	infinite	slightly	insoluble
Ethers, aldehydes, ketones	C3: slightly	C4+: insoluble	insoluble
Hydrocarbons (ALL, incl. benzene), thiols	insoluble	insoluble	insoluble
Phenols	insoluble	insoluble	insoluble

Phenol + NaOH gives soluble sodium phenoxide. Amine drugs + HCl give soluble ammonium salts. Essay trick: 'which hydrocarbon is more soluble?' Neither.

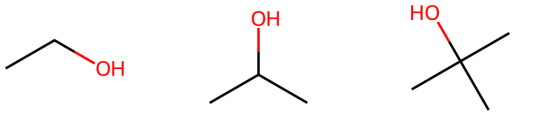
Solubility boundaries

- Her ranking steps: **(1) IMF** (H-bond > dipole > London), **(2) size** (more carbons = higher), **(3) bp:** branching lowers it. Melting point also depends on **packing** ("better packing").
- Impurities **lower** the melting point and **raise** the boiling point.
- ethane bp is **−89 °C**; methanamine −6 °C; methanol 65 °C; diethyl ether 35 °C; 1-butanol 118 °C.

ALCOHOLS

- **–OH on an sp³ carbon.** Class = carbons on the C–OH carbon.

Alcohol class = carbons on the C–OH carbon. α-H = H on the C–OH carbon.



1° (ethanol)
2 α-H

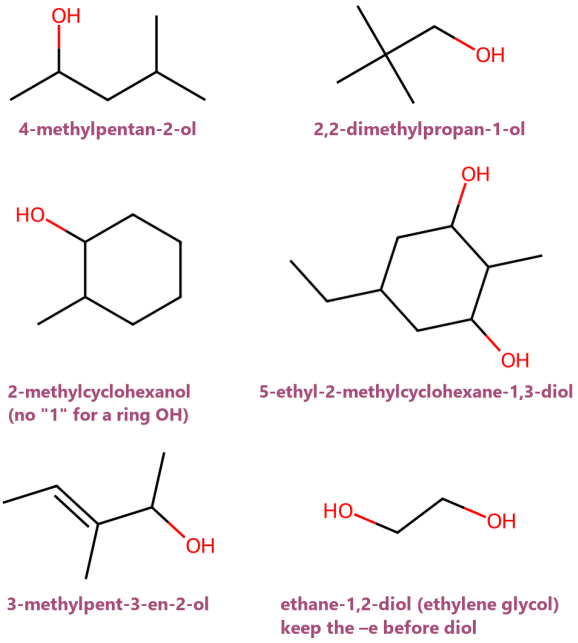
2° (2-propanol)
1 α-H

3° (2-methyl-2-propanol)
0 α-H

Class and α-hydrogens decide oxidation

NAMING

- Longest chain that **contains C–OH**; number from the **OH end**; drop **-e**, add **-ol** (keep **-an-**): **propan-2-ol**. Locate the OH from C3 up.
- OH beats C=C for the low number: **3-methylpent-3-en-2-ol** (**-en-** before **-ol**).
- Ring OH is C1, **no "1"**: **2-methylcyclohexanol**.
- Diols/triols **keep the -e**: **ethane-1,2-diol** (glycol), **propane-1,2,3-triol** (glycerol).
- Common: isopropyl alcohol, sec-butyl, tert-butyl alcohol.

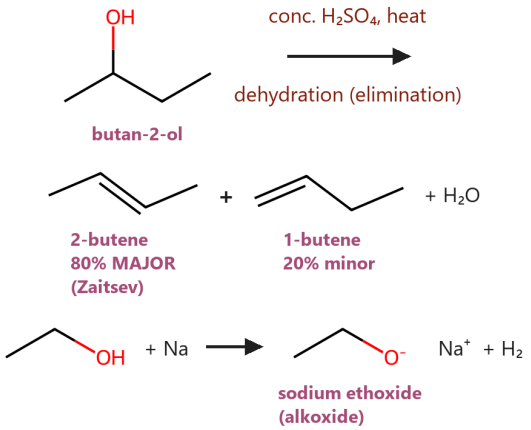


Her naming drills

REACTIONS

- Alcohols are **neutral** (no reaction with NaOH, unlike phenol). With **Na metal**: **alkoxide + H_2** (not NaOH).
- **Dehydration:** conc. H_2SO_4 + heat gives **alkene + H_2O** (elimination). **Zaitsev:** the **more substituted alkene** is major. Needs 2 carbons (methanol cannot).

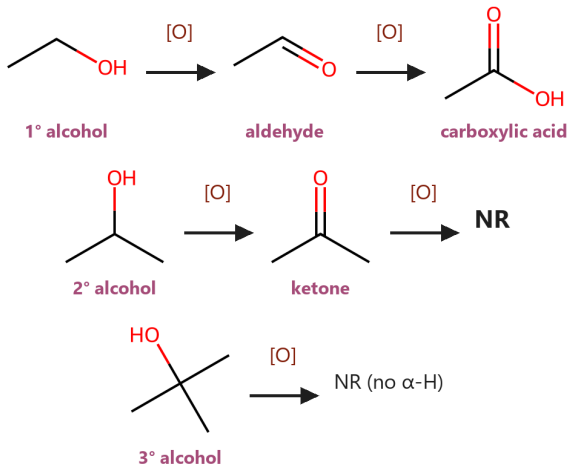
Ease: 3° (50 °C) > 2° (140 °C) > 1° (180 °C). Needs 2+ carbons. Zaitsev: the H leaves the carbon with FEWER H's.



Zaitsev dehydration, and alcohol + Na

- **Oxidation:** 1° gives aldehyde then **acid**; 2° gives **ketone** (stops); 3° **NR**.

[O] = $KMnO_4$, $K_2Cr_2O_7$ / H_2SO_4 or PCC (PCC stops 1° at the aldehyde). In the body: alcohol dehydrogenase.



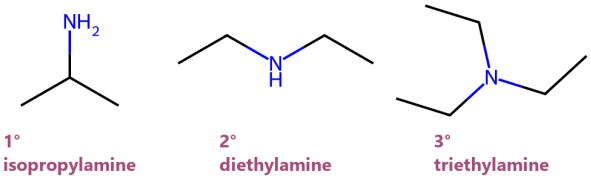
Oxidation by class

- In the body (**alcohol dehydrogenase**): ethanol gives **ethanal** (hangover) then acetic acid; 2-propanol gives **acetone**; **methanol gives methanal then formic acid** (blindness, lambanog).

AMINES

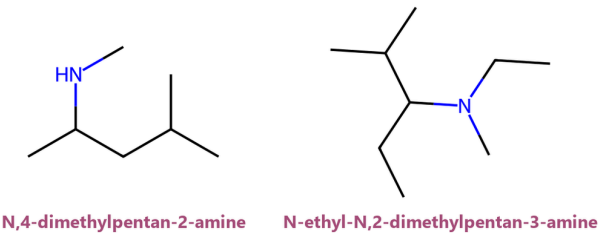
- Class = carbons **on the N** (opposite of alcohols' trap). **Aromatic amine**: N on a ring (aniline). **Heterocyclic**: N in the ring (pyridine, pyrimidine, purine).

Amine class = carbons ON THE NITROGEN (isopropylamine is 1°, even though its carbon is 2°).



Count the carbons on nitrogen

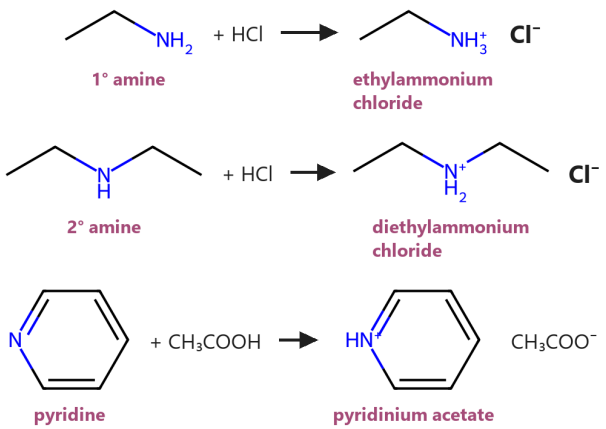
- IUPAC: drop **-e**, add **-amine** (**propan-2-amine**). Groups on N get the locant **N-**. Common name: groups alphabetical + amine.



N- locants

- Smells: trimethylamine (fish), **putrescine** (rotting meat), **cadaverine**.
- 1° and 2° amines H-bond (N–H···N), 3° cannot, but all H-bond with water. bp **below alcohols**, above ethers.
- **Amines are bases** (N lone pair). **Amine + acid = ammonium salt, no water.** Salts are water-soluble: why drugs are made as HCl salts.

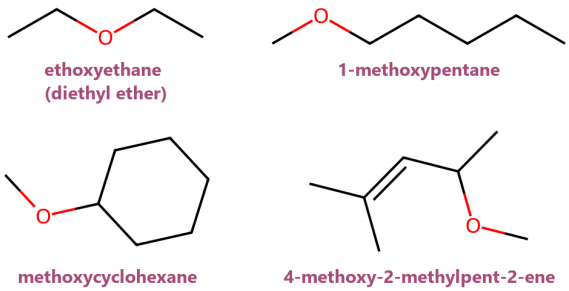
"Walay water!" Copy the amine, add one H to N, make N^+, add the anion. A salt only.



Copy the amine, add H to N, make it +, add the anion

ETHERS

- **R–O–R'**. **Always a branch (alkoxy-)**: longer chain is the parent, shorter group becomes **methoxy-, ethoxy-, propoxy-**.
- No O–H, **no H-bonding** between molecules: bp far below alcohols (35 vs 118 °C). **C3 slightly soluble, C4+ insoluble.**



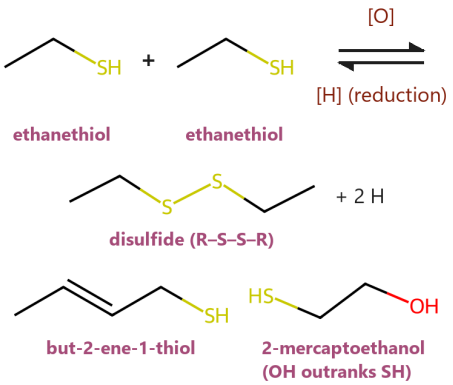
Ethers are named as alkoxy branches

- C=C gets the lower number than the alkoxy group.
- Cyclic: **ethylene oxide** (sterilant), **THF** (solvent). Ethers are unreactive.

THIOLS

- **–SH**; **keep the -e**: **ethanethiol**. With a higher group: **mercapto-** (2-mercaptoethanol). Common: **mercaptan**.
- S–H nearly nonpolar: **insoluble**, bp only a bit above hydrocarbons (methanethiol 6 °C vs methanol 65 °C).
- **Stench:** ethanethiol is added to **LPG** so you smell leaks. Onion, durian, skunk.
- Weak acid (pK_a about 10): NaOH gives **thiolate** (RS^-).

Hair rebonding: reducing agent breaks cysteine –S–S–, then an oxidizing agent re-forms it.



Thiol and disulfide: oxidation and reduction

- **Cysteine** –S–S– cross-links in **keratin**. Rebonding: reduce, straighten, oxidize.

END OF LESSON

Topic 14: Aldehydes and Ketones

The carbonyl families

CHEMISTRY (PROF. L. RANAY) | LECTURE QUIZ, NOV 7: TOPICS 12-14

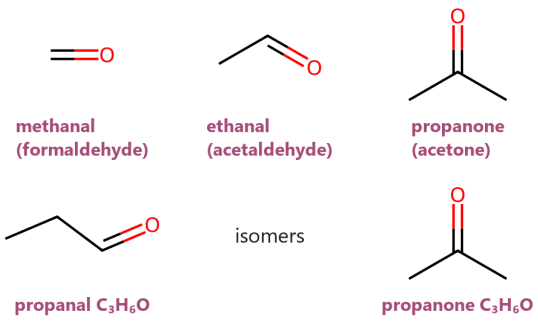
What she stressed for the quiz

Priority order and oxo/hydroxy/amino/mercapto. **CHO is always C1.** Ring CHO = **carbaldehyde**. **Tollens / Benedict**: aldehyde yes, ketone no. **Hemiacetal vs acetal** needs an alcohol. **Two enols** for each ketone. bp: higher than alkanes and ethers, **lower than alcohols**.

STRUCTURE

- **C=O is polar** (O δ−, C δ+). **Aldehyde:** carbonyl C has **at least one H** (smallest: **methanal**). **Ketone:** carbonyl C between **two carbons** (smallest: **propanone**).
- Same carbon count: **functional group isomers** (propanal and propanone are both C_3H_6O).

Aldehyde: C=O carbon has at least one H. Ketone: C=O carbon between two carbons (smallest = C3).



Aldehydes and ketones

NAMING

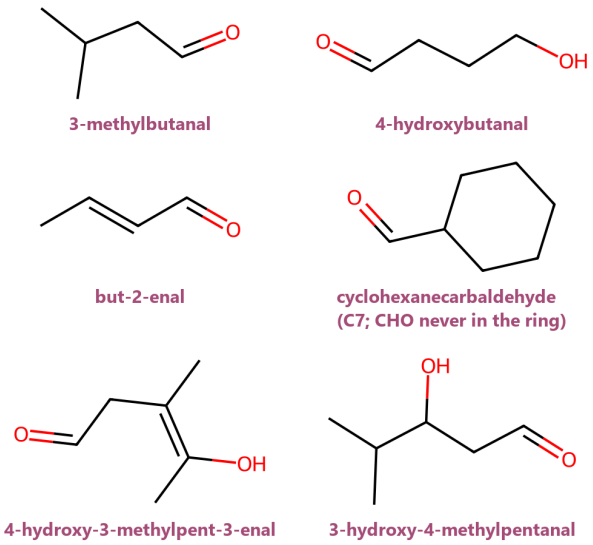
NAMING PRIORITY (highest to lowest)

Group	As parent	As a branch
1. Carboxylic acid	-oic acid	never a branch
2. Aldehyde	-al	oxo-
3. Ketone	-one	oxo-
4. Alcohol	-ol	hydroxy-
5. Amine	-amine	amino-
6. Thiol	-thiol	mercapto-

Ethers, halides, nitro are always branches (alkoxy-, chloro-, nitro-). Then: double bond, nearest branch, alphabetical.

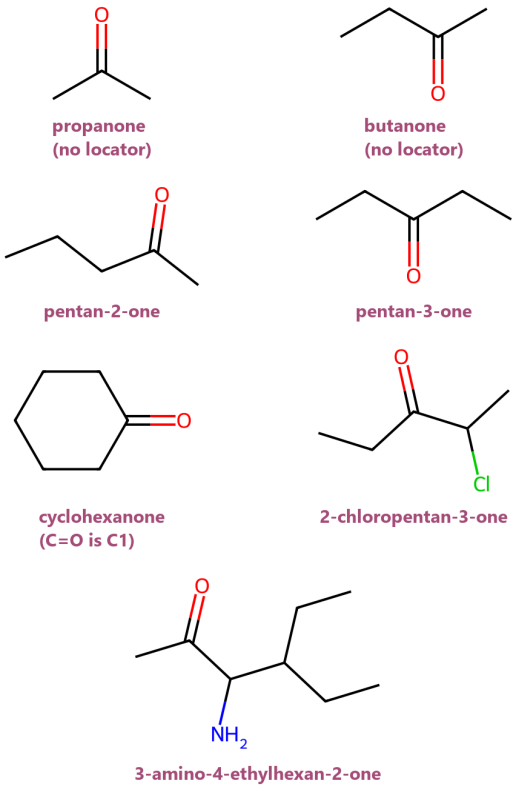
Priority order

- Aldehyde beats ketone, so a molecule with both is named as the aldehyde (C=O as **oxo-**).
- **Aldehyde:** drop -e, add **-al**. CHO is **C1**, no locator. **Ring:** a CHO carbon cannot be in the ring (5 bonds), so ring name + **carbaldehyde**. C=C: **but-2-enal**.



Aldehyde naming

- **Ketone:** drop -e, add **-one**. **Propanone and butanone: no locator.** From **pentanone up**, locate it. Ring ketone C=O is C1. Numbering: group, double bond, branch, alphabetical.



Ketone naming

Note: Her butanone shortcuts (like "2-methylbutanone") are not quizzed here. Use current IUPAC if you must name one: 3-methylbutan-2-one.

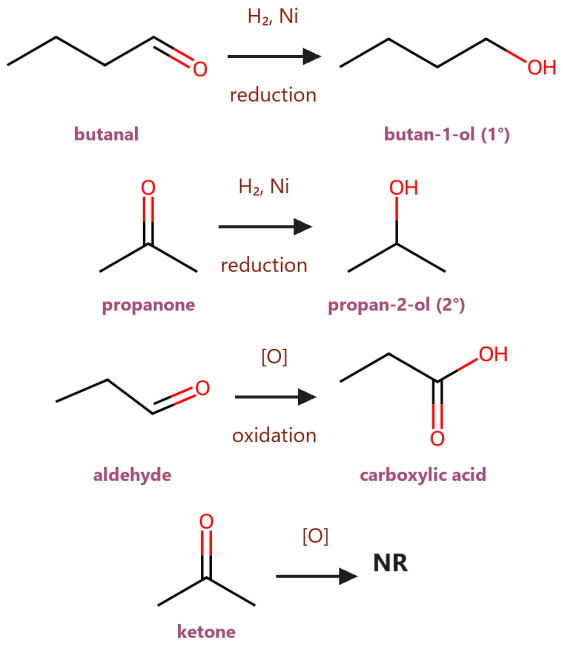
PHYSICAL PROPERTIES

- **Dipole-dipole only, no H-bonding.** bp: **above alkanes and ethers, below alcohols and acids.** Her table: diethyl ether 34 < pentane 36 < butanal 76 < butanone 80 < 1-butanol 117 < propanoic acid 141 °C.
- Branching lowers bp. **Solubility: C3 slightly, C4+ insoluble** (same as ethers).
- Full ladder and solubility table: see Topic 13 boxes.

OXIDATION AND REDUCTION

- **Aldehydes oxidize** (to acids), **ketones resist** (an aldehyde has an **H on the carbonyl C**). Reagents: **$K_2Cr_2O_7$, $KMnO_4$, PCC**.
- 1° alcohol, aldehyde, acid. 2° alcohol, ketone, then **NR**. 3° **NR**. **Reduction is the reverse.**

Aldehyde to 1° alcohol, ketone to 2° alcohol. Body: NADH reduces (pyruvate to lactate).

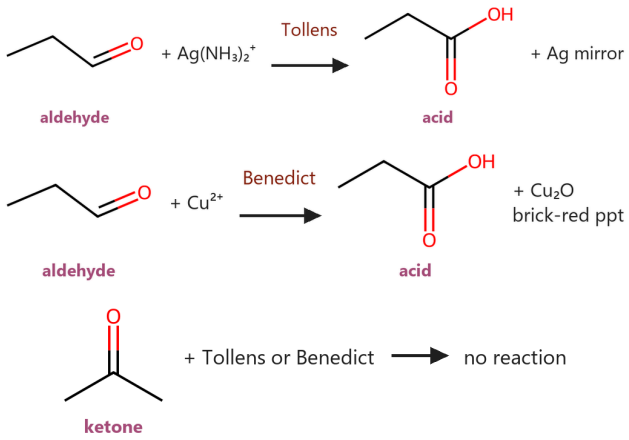


Reduction and oxidation

TOLLENS AND BENEDICT

- **Tollens** (ammoniacal silver nitrate): **silver mirror** = aldehyde. **Benedict / Fehling** ($CuSO_4$): **brick-red** = aldehyde (sugar in urine). Ketones: no change.

Sugars are aldehydes: brick-red Benedict = sugar in urine.



Tests that tell them apart

- Body: **NADH** reduces (pyruvate to lactate). Cortisol has a ketone; aldosterone has an aldehyde and a ketone.

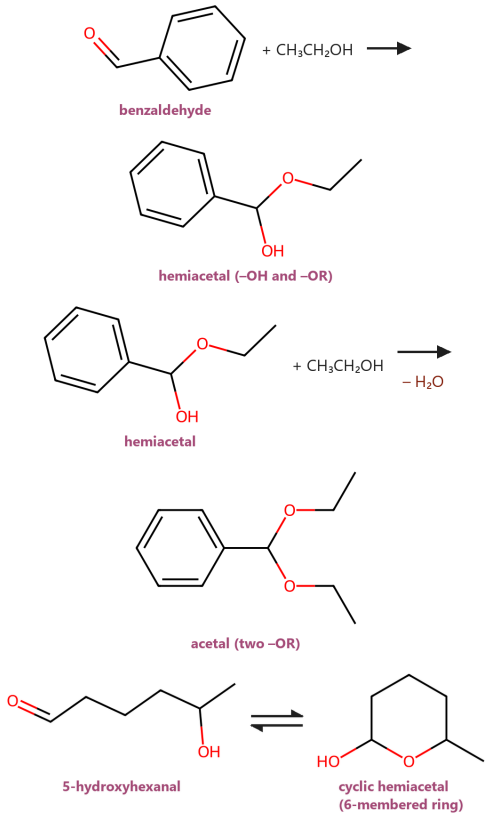
HEMIACETALS AND ACETALS

- The **C=O π bond** adds an alcohol (aldehydes and ketones both). **Hemiacetal:** H to the carbonyl O,

OR to the C (one –OH, one –OR). **Acetal:** second alcohol, **lose H_2O**, two –OR.

- **Cyclic hemiacetal:** an OH in the same molecule closes a **5- or 6-ring** (5-hydroxyhexanal).

Needs a carbonyl plus an alcohol ($KMnO_4$ instead means oxidation).
Hemiacetal: H to the O, OR to the C.

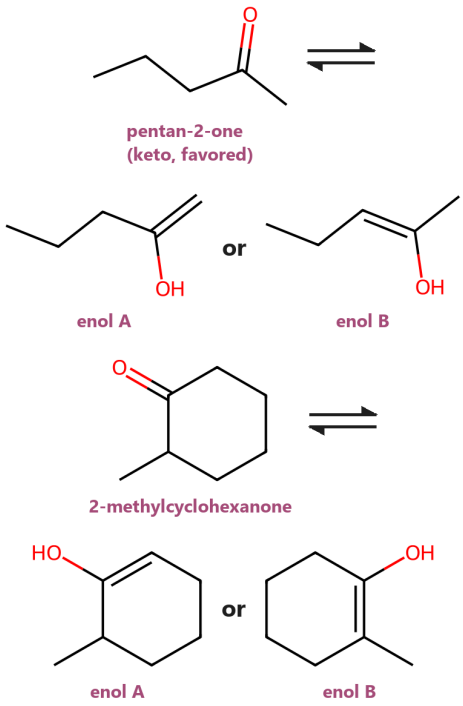


Hemiacetal, acetal, cyclic hemiacetal

KETO-ENOL TAUTOMERISM

- Needs a **C=O with an α-hydrogen**. H moves to O, C=O becomes C=C: **ketone ⇌ enol**. The **keto form predominates**.

Needs an α-hydrogen: H moves to O, C=O becomes C=C–OH. Each of these ketones has TWO enols.



Two enols each

END OF LESSON